



An Explainable AI Multi-Agent Recommender System for Financial Document Access Control

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Abstract

Financial institutions require robust document access control mechanisms that balance security with transparency and explainability. Traditional classification systems often operate as black boxes, failing to provide justifications for access-control decisions. This work presents a novel explainable AI multi-agent recommender system for financial document sensitivity classification that addresses critical ethical concerns in AI-powered decision-making. We fine-tuned three state-of-the-art models—FinBERT, BERT-base-uncased, and GPT-4.1-mini—on a custom-labeled Financial PhraseBank dataset with four sensitivity levels: Public, Internal, Confidential, and Restricted. These fine-tuned models serve as specialized AI agents within a multi-agent architecture orchestrated by GPT-5.1, a large reasoning model operating in zero-shot mode. The orchestrator synthesizes agent predictions and generates natural language recommendations that justify classification decisions. Our agentic AI multi-agent recommender system achieves 83.71% overall accuracy, comparable to individual models (82.80%–84.93%), while providing interpretable explanations for each decision. Critically, agent agreement analysis reveals that unanimous decisions (3/3 agents agree, 78.8% of cases) achieve 92.28% accuracy—significantly outperforming any individual model—validating the collaborative decision-making approach. The system demonstrates that multi-agent architectures can provide both high-confidence predictions and natural language explainability, creating transparent, accountable AI systems for financial document access control. All code and methodologies are released as open-source on our GitHub (Applied-AI-Research-Lab 2025) to support reproducibility and further research in explainable AI for finance.

Keywords Document access control · Recommender systems · Multi-agent system · Explainable AI · Decision support systems · Business intelligence

1 Introduction

Financial institutions manage vast repositories of sensitive documents containing market-moving information, strategic plans, confidential analyses, and material non-public data (Li et al., 2016; Uddin et al., 2019). Proper classification and access control of these documents is crucial for regulatory compliance, competitive advantage protection, and risk management. However, traditional document classification systems face a fundamental challenge: they operate as black boxes that provide classifications without explanations, making it impossible for stakeholders to understand, trust, or validate access-control decisions (Liu et al., 2025; Shao et al., 2025). This opacity becomes particularly problematic in the financial sector, where incorrect classifications can lead to severe consequences including regulatory violations, insider trading accusations, competitive intelligence

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leaks, or unauthorized disclosure of material information (Chiu & Hung, 2005). Furthermore, as artificial intelligence systems increasingly make critical decisions about data access, the need for transparency, accountability, and explainability intensifies—especially given growing regulatory requirements such as the European Union’s AI Act and various financial compliance frameworks (Hussain & Hussain, 2025; Jo & Raj, 2024; Visave, 2025). The imperative for transparent, resilient information management systems has been further amplified by recent crises, during which organizations faced unprecedented challenges in managing misinformation, maintaining operational continuity, and ensuring trusted access to critical information resources (Spanaki et al., 2024). Research in the GRC domain similarly shows that effective compliance-oriented systems depend on clearly articulated objectives, stakeholder roles, and implementation requirements, underscoring the importance of transparency in complex risk-management environments (Papazafeiropoulou & Spanaki, 2016). Moreover, investigations into organizational responses to cybersecurity incidents reveal that employee emotional reactions and coping mechanisms significantly influence the effectiveness of information security measures, highlighting the human dimension of access control systems (Stacey et al., 2021). This challenge is not unique to financial institutions; related work has shown that argumentation-based and abductive-reasoning frameworks grounded in data-quality principles can support more transparent and controlled data processing (Karafili et al., 2019). Advanced frameworks for cybersecurity governance, particularly in complex supply chain environments, demonstrate that regulatory compliance requires not only technical controls but also comprehensive risk assessment methodologies and adaptive security architectures (Gokkaya et al., 2025).

A fundamental challenge in developing AI systems for financial document access control is the complete absence of publicly available training datasets. Unlike other NLP domains where benchmarks proliferate, financial document sensitivity classification faces an insurmountable privacy barrier: real-world sensitive documents cannot be shared due to regulatory constraints (securities law, trade secrets), competitive risks (strategic information disclosure), and legal liability (breach of fiduciary duty) (Chiu & Hung, 2005; Papazafeiropoulou & Spanaki, 2016). This creates a paradox: organizations need trained models to implement automated access control, yet they cannot share the data required to develop these models. Each financial institution must train models independently using proprietary data, preventing research advancement, reproducibility validation, and collaborative improvement—core tenets of scientific progress. Similar challenges pervade broader information management contexts where sensitive data

sharing must balance utility with security, as evidenced by AI-driven frameworks for data access control in Industry 4.0 environments where role-based policies govern multi-stakeholder information flows (Spanaki et al., 2021). To address this gap, we adopt a synthetic dataset generation approach, transforming the publicly available Financial PhraseBank sentiment dataset (Malo et al., 2014) into a sensitivity classification dataset through systematic keyword-based heuristics and sentiment-aware label mapping. This methodology aligns with established practices in privacy-sensitive domains where synthetic data generation, data augmentation, and label transformation enable research when real data cannot be disclosed (Jordon et al., 2022; Chen et al., 2021; Xu et al., 2019). Recent advances demonstrate the validity of synthetic datasets across multiple domains: in healthcare, synthetic patient records enable HIPAA-compliant research (Tucker et al., 2020; Yoon et al., 2020); in finance, synthetic transaction data supports fraud detection research without exposing customer information (Lopez-Rojas & Axelsson, 2022; Rankin et al., 2022); and in NLP, data augmentation and label transformation techniques successfully extend limited training data (Wei & Zou, 2019; Karimi et al., 2021; Kumar et al., 2021). Our approach builds on these precedents, explicitly acknowledging that our synthetic labels serve as a *proxy task* for demonstrating the multi-agent system’s capabilities rather than claiming perfect real-world fidelity.

The financial industry faces a paradox: while AI-powered document classification can enhance efficiency and security, the lack of explainability creates risks that may outweigh these benefits. Financial professionals need to understand *why* a document receives a particular sensitivity classification to validate that the classification aligns with regulatory requirements and internal policies, identify and correct misclassifications before they cause harm, provide audit trails for compliance and governance, build trust in automated systems among human stakeholders, and enable informed override decisions when automated classifications appear incorrect. The importance of explainability extends beyond technical accuracy to encompass organizational trust and resilience, particularly as digital technologies increasingly mediate decision-making in entrepreneurial and growth-oriented financial firms where information asymmetries can constrain innovation (Balta et al., 2023). Recent advances in large language models (LLMs) have opened new possibilities for explainable AI, as these models can generate natural language justifications for their predictions (Černevičienė & Kabašinskas, 2024; Saarela & Podgorelec, 2024). Research on explainable AI in B2B service contexts demonstrates that transparency mechanisms can enhance organizational resilience by enabling stakeholders to validate system recommendations and adapt to dynamic business environments

(Olan et al., 2026). However, LLMs also have limitations including hallucination (Sriramanan et al., 2024; Banerjee et al., 2025), inconsistency, and domain-specific knowledge gaps (Zhang et al., 2024; Welz & Lanquillon, 2024). Meanwhile, domain-specific models like FinBERT (Araci, 2019; ProsusAI, Hugging Face, 2026) excel at financial text understanding but lack the reasoning and explanation generation capabilities of LLMs (Nasiopoulos et al., 2025). Case studies of FinTech project implementations reveal that success often hinges not merely on technical sophistication but on the ability to provide stakeholders with transparent, comprehensible rationales for system decisions (Jinasena et al., 2023).

Despite advances in financial NLP, explainable AI, and multi-agent systems, a significant gap remains: existing approaches either prioritize accuracy without explainability (black-box models), provide limited explanations that don't support decision validation (attention mechanisms, importance scores), or lack the domain-specific understanding required for financial applications. We hypothesize that a multi-agent system combining the strengths of different model architectures—domain-specific transformers, general-purpose language models, and advanced reasoning models—can achieve superior performance and explainability compared to any single model approach. Specifically, we propose that fine-tuned specialist models can provide reliable domain-specific predictions, an advanced reasoning model (orchestrator) can synthesize multiple expert opinions, the collaborative decision-making process can generate more accurate and trustworthy classifications, natural language explanations from the orchestrator can justify decisions transparently, and synthetic dataset generation provides a valid methodology for developing and evaluating access control systems in privacy-constrained domains. The primary objectives of this research are to develop and validate a systematic synthetic sensitivity labeling methodology for financial texts that transforms sentiment-annotated data into access control classifications, fine-tune and evaluate multiple model architectures on financial document sensitivity classification, design and implement a multi-agent agentic AI system governed by an intelligent orchestrator, generate interpretable recommendations and explanations for classification decisions, compare the agentic system's overall performance against individual model approaches, analyze the collaborative decision-making dynamics within the multi-agent architecture, and demonstrate that synthetic datasets can serve as effective proxies for real-world tasks when properly constructed and transparently acknowledged. Our work addresses this critical gap by combining fine-tuned domain-specific and general-purpose models as specialist agents with an advanced reasoning model as orchestrator. This architecture enables both high accuracy

through collaborative decision-making and natural language explainability through the orchestrator's reasoning capabilities. Furthermore, by focusing on financial document sensitivity classification for access control, we address a practical application area with significant real-world impact and regulatory importance.

This work makes the following contributions to the fields of explainable AI, financial technology, and document access control:

1. **Novel Dataset Construction:** We develop a methodology for creating synthetic sensitivity labels on the Financial PhraseBank dataset (Malo et al., 2014), mapping financial sentiment to four-level security classifications through keyword-based heuristics and sentiment analysis.
2. **Comprehensive Model Comparison:** We fine-tune and rigorously evaluate three distinct model architectures—FinBERT (domain-specific) (Araci, 2019; ProsusAI, Hugging Face, 2026), BERT-base-uncased (general NLP) (Devlin et al., 2018), and GPT-4.1-mini (LLM) (OpenAI, 2025)—demonstrating that each achieves 83-85% individual accuracy on sensitivity classification.
3. **Agentic AI Multi-Agent Architecture:** We design a multi-agent system where three fine-tuned specialist AI agents provide independent assessments, and a GPT-5.1 (OpenAI, 2025) orchestrator (operating in a zero-shot mode) synthesizes their predictions through advanced reasoning to make final decisions.
4. **Explainable Recommendations with Confidence Calibration:** The orchestrator generates natural language explanations for each classification. When all three agents unanimously agree (78.8% of cases), the system achieves 92.28% accuracy—significantly outperforming individual models—providing both transparency and reliable confidence indicators for automated decisions.
5. **Open-Source Contribution:** We release the complete codebase, including data preparation, model fine-tuning, agentic AI system implementation, and evaluation scripts, to support reproducibility and future research (Applied-AI-Research-Lab, 2025).

The remainder of this paper is structured as follows: Section 2 reviews related literature on financial NLP, explainable AI, and multi-agent systems. Section 3 describes our methodology, including dataset preparation, model fine-tuning, and agentic system architecture. Section 4 presents experimental results and performance analysis. Section 5 discusses implications, limitations, and future directions.

Section 6 concludes the paper with key findings and contributions.

2 Related Work

2.1 Financial Natural Language Processing

Financial NLP has emerged as a critical research area due to the unique characteristics of financial texts, including specialized vocabulary, temporal sensitivity, and the high stakes associated with interpretation errors. Early approaches relied on dictionary-based sentiment analysis and rule-based systems, but modern deep learning methods have significantly improved performance on financial text understanding tasks.

The Financial PhraseBank dataset (Malo et al., 2014) has become a benchmark for financial sentiment analysis, containing 4,840 sentences from English-language financial news annotated by multiple experts. While originally designed for three-class sentiment classification (positive, neutral, negative), its high-quality annotations and domain specificity make it valuable for other financial text understanding tasks.

FinBERT (Araci, 2019; ProsusAI, Hugging Face, 2026), a BERT model pre-trained on financial communication texts, demonstrated that domain-specific pre-training significantly improves performance on financial NLP tasks. Subsequent works have explored various financial applications including stock price prediction (Chen et al., 2025; Kim et al., 2023), risk assessment (Taheripour et al., 2025a, b), and regulatory document analysis (Huang et al., 2023). However, most research has focused on classification accuracy rather than explainability and access control applications.

2.2 Document Classification and Access Control

Document access control traditionally relies on manual classification processes or rule-based systems that apply keyword matching and metadata analysis (Anutariya et al., 2003; Hou & Lin, 2004; Lu & Sagduyu, 2016). While these approaches offer transparency, they struggle with scale, consistency, and the nuanced understanding required for modern document repositories.

Machine learning approaches to document classification have evolved from traditional methods like Naive Bayes (Liu et al., 2017; Sathyadevan et al., 2014) and SVM (Thielmann et al., 2023) to deep learning architectures including CNNs (Liu et al., 2019), RNNs (Rafi et al., 2025; Saxena & Sun, 2024), and transformers (Toprak & Turan, 2024). Recent comparative studies have evaluated machine

learning versus deep learning techniques for text analytics tasks, demonstrating that model selection significantly impacts classification performance across different text domains (Marshan et al., 2025). These findings underscore the importance of architectural choices in text classification systems, particularly when dealing with nuanced categorization tasks. However, applying these models to security-sensitive domains like financial document access control introduces unique challenges: classifications must be both accurate and explainable, errors can have severe consequences, and domain expertise is essential for validation. In broader information systems contexts, the implementation of governance, risk, and compliance (GRC) systems has been shown to require careful consideration of integration, optimization, and information value drivers throughout the entire system lifecycle (Spanaki & Papazafeiropoulou, 2016), principles that extend naturally to AI-powered access control implementations in financial organizations.

2.3 Explainable AI in Finance

Explainable AI (XAI) has gained significant attention as AI systems increasingly influence high-stakes decisions. In finance, regulators and practitioners demand transparency to ensure fairness, detect bias, enable audit trails, and maintain trust (Weber et al., 2024; Yeo et al., 2025). Research on digital transformation strategies emphasizes that successful AI adoption in organizations requires balancing technological capabilities with stakeholder trust and value creation, particularly in dynamic business environments where transparency becomes a competitive differentiator (Spanaki et al., 2024). Traditional XAI approaches include attention visualization, feature importance scores (e.g., LIME (Garreau & Luxburg, 2020), SHAP (Lin & Gao, 2022)), and rule extraction from neural networks (Černevičienė & Kabašinskas, 2024).

LLMs have introduced new possibilities for explainability through natural language generation. Rather than providing numeric importance scores or visualizations, LLMs can generate human-readable explanations that describe reasoning processes (Roumeliotis et al., 2025). This approach aligns naturally with human communication but introduces challenges including hallucination (Banerjee et al., 2025; Sriramanan et al., 2024), post-hoc rationalization, and consistency issues.

2.4 Multi-Agent Systems and Agentic AI

Multi-agent systems have a long history in AI, enabling complex problem-solving through collaboration among specialized agents. Traditional approaches often involved

rule-based coordination or game-theoretic mechanisms. Recent advances in LLMs have enabled new forms of agentic AI where LLM-powered agents can reason, communicate, and collaborate using natural language or machine-native encodings (Acharya et al., 2025; Sapkota et al., 2026).

Research on LLM-based multi-agent systems has explored various architectures including debate frameworks where agents argue different positions, ensemble methods where predictions are aggregated, and hierarchical systems where a coordinator synthesizes agent inputs (Hosseini & Seilani, 2025; Roumeliotis et al., 2026). These approaches have shown promise in domains including code generation, complex reasoning tasks, and question answering.

However, no studies have applied agentic AI to the finance sector, particularly for document classification or access control. The combination of domain-specific specialist models with general-purpose reasoning capabilities represents a novel approach that leverages the complementary strengths of different model architectures.

2.5 Synthetic Data Generation and Label Transformation

The use of synthetic data for training machine learning models has gained substantial traction in privacy-sensitive domains where real data cannot be shared. Synthetic data generation encompasses multiple approaches including generative models that create entirely artificial samples, data augmentation techniques that transform existing data, and label transformation methods that repurpose annotated datasets for related tasks (Jordon et al., 2022).

In healthcare, synthetic patient data enables research without HIPAA violations. Studies demonstrate that generative adversarial networks (GANs) and variational autoencoders (VAEs) can produce synthetic electronic health records maintaining statistical properties while protecting privacy (Tucker et al., 2020; Yoon et al., 2020). Similarly, in finance, synthetic transaction datasets support fraud detection and risk modeling research (Lopez-Rojas & Axelsson, 2022; Rankin et al., 2022).

For NLP tasks, data augmentation has become standard practice when training data is limited. Techniques include back-translation, synonym replacement, random insertion and deletion (EDA) (Wei & Zou, 2019), contextual word embeddings substitution (AEDA) (Karimi et al., 2021), and paraphrasing using language models (Kumar et al., 2021). Advanced text analytics research has demonstrated that extracting comprehensive textual features—including brevity, readability, sentiment, and linguistic patterns—significantly enhances classification model performance across diverse application domains (Dousios et al., 2025). These methods demonstrably improve model robustness and

generalization, particularly for low-resource languages and specialized domains.

Label transformation—repurposing datasets annotated for one task to train models for a related task—represents another established approach. For example, sentiment analysis datasets have been successfully adapted for emotion detection (Mohammad, 2018), stance detection (Küçük & Can, 2020), and opinion mining (Pontiki et al., 2016). The key requirement is that the source and target tasks share underlying linguistic or semantic relationships that models can learn to exploit.

Our synthetic labeling methodology follows this tradition: we transform sentiment-annotated financial texts into sensitivity classifications through systematic heuristics that map content patterns (keywords, sentiment polarity) to access control levels. While our synthetic labels do not represent ground-truth sensitivity decisions from real financial institutions, they create a *consistent, learnable task* that enables us to develop and evaluate our multi-agent system architecture. This approach is methodologically valid provided we (a) transparently acknowledge the synthetic nature of labels, (b) use systematic, reproducible labeling rules, and (c) ultimately validate on real-world data when feasible (Chen et al., 2021; Xu et al., 2019).

3 Methods

3.1 Dataset Selection and Preparation

3.1.1 Financial PhraseBank Dataset

We selected the Financial PhraseBank dataset (Malo et al., 2014) as the foundation for our study for several key reasons. First, the dataset contains 4,840 sentences drawn from English-language financial news, providing content that is highly relevant to real-world financial communication. In addition, it includes the `Sentences_AllAgree.txt` subset, in which all annotators unanimously agreed on the sentiment labels. This consensus ensures high-quality annotations and minimizes noise when deriving our sensitivity classifications. Finally, the dataset's sentences closely resemble the types of communications commonly encountered by financial institutions, further supporting the practical relevance of our work.

While the original dataset assigns sentences to three sentiment categories (positive, neutral, and negative), our task requires four levels of sensitivity: Public, Internal, Confidential, and Restricted. To bridge this gap, we developed a novel methodology to infer sensitivity labels from the existing annotations while preserving both the

domain specificity and the high annotation quality of the original corpus.

3.1.2 Synthetic Sensitivity Label Generation

Since no existing dataset provides sensitivity classifications for financial documents, we created a systematic approach to generate synthetic labels that reflect realistic document access control requirements. Our methodology combines keyword-based heuristics with sentiment-aware adjustments to assign sensitivity levels.

We defined four comprehensive keyword dictionaries corresponding to each sensitivity level. **Restricted (Level 3)** keywords indicate material non-public information or regulatory concerns such as “insider,” “confidential,” “unreleased,” “merger,” “acquisition,” “regulatory,” “SEC filing,” “material information,” “non-public,” and “restricted.” **Confidential (Level 2)** keywords suggest strategic internal information including “strategic,” “competitive,” “proprietary,” “internal,” “forecast,” “projection,” “executive,” “board,” “management decision,” and “restructuring.” **Internal (Level 1)** keywords indicate company-specific operational information such as “company,” “analysis,” “performance,” “revenue,” “profit,” “loss,” “growth,” “market share,” “competitor,” and “industry.” Finally, **Public (Level 0)** keywords suggest already-disclosed information including “public,” “announced,” “reported,” “statement,” “press release,” “market,” “general,” “news,” “sector,” and “economic.”

The labeling algorithm proceeds in five steps:

1. Normalize the text by converting all tokens to lower-case to enable case-insensitive matching.
2. Check for keyword matches beginning with the highest sensitivity level and proceeding downward.
3. Assign the highest sensitivity level for which at least one keyword is detected.
4. Apply a sentiment-based adjustment: if the sentiment label is negative and the assigned base level is less than 3, increase the sensitivity level by one. This adjustment reflects the tendency for negative disclosures to require stricter access control.
5. Handle sentences with no keyword matches by assigning probabilistic defaults based on sentiment: negative sentences are assigned to Internal (70%) or Confidential (30%), while neutral or positive sentences are assigned to Public (60%) or Internal (40%).

This approach ensures that every sentence receives a sensitivity label grounded in both content keywords and sentiment polarity, creating a realistic distribution of sensitivity levels.

3.1.3 Class Balancing through Oversampling

Initial labeling produced an imbalanced class distribution due to the keyword-based heuristic favoring certain sensitivity levels. To ensure models learn equally from all classes and prevent bias toward majority classes, we applied stratified oversampling with replacement.

We identified the maximum class size and upsampled all minority classes to match this size through random sampling with replacement. This technique, while introducing some duplicate samples, ensures balanced representation during training and prevents models from defaulting to majority-class predictions. The balancing process transformed the dataset from its original imbalanced state to a fully balanced distribution with equal representation across all four sensitivity levels.

3.1.4 Train-Validation-Test Split

We performed stratified splitting to create three subsets while maintaining class balance. The **training set** comprises 70% of the balanced data for model fine-tuning, the **validation set** contains 15% for hyperparameter tuning and early stopping, and the **test set** holds 15% for final evaluation (completely held out until all models were trained). Stratification ensured that each subset contains proportional representation of all four sensitivity levels, preventing evaluation bias and enabling fair comparison across models. The test set remained completely isolated during model development to provide unbiased performance estimates.

Table 1 summarizes the dataset statistics. Figure 1 illustrates the train-validation-test split proportions and the effect of oversampling on class distribution.

3.2 Model Fine-Tuning Methodology

We fine-tuned three distinct model architectures to serve as specialist AI agents in our multi-agent system. Each model brings unique strengths: domain-specific knowledge (FinBERT), general language understanding (BERT-base-uncased), and large-scale reasoning capabilities (GPT-4.1-mini).

The selection of this specific model ensemble was strategically designed to test distinct hypotheses about the value of architectural diversity in multi-agent systems for financial document classification. We chose FinBERT as the domain-specialist agent because it represents the state-of-the-art in financial NLP, having been pre-trained on extensive financial corpora including earnings reports, analyst communications, and financial news (Araci, 2019; ProsusAI, Hugging Face, 2026). This enables direct evaluation of whether domain-specific pre-training confers advantages

Table 1 Comprehensive statistics of the Financial PhraseBank dataset transformation for sensitivity classification. The table shows the original dataset characteristics, processed dataset with synthetic sensitivity labels, balanced dataset after oversampling, and final train-validation-test split proportions

Metric	Value
Original Dataset	
Total Sentences	2,264
Average Words per Sentence	22.4
Average Characters per Sentence	122.0
Min/Max Words	2/81
Processed Dataset	
Total Samples	2,264
Negative Sentiment	303
Neutral Sentiment	1,391
Positive Sentiment	570
Public Sensitivity	838
Internal Sensitivity	1,094
Confidential Sensitivity	298
Restricted Sensitivity	34
Balanced Dataset (Oversampled)	
Total Samples	4,376
Samples per Sensitivity Level	1,094
Train/Val/Test Split	
Training Set (70%)	3,063
Validation Set (15%)	656
Test Set (15%)	657

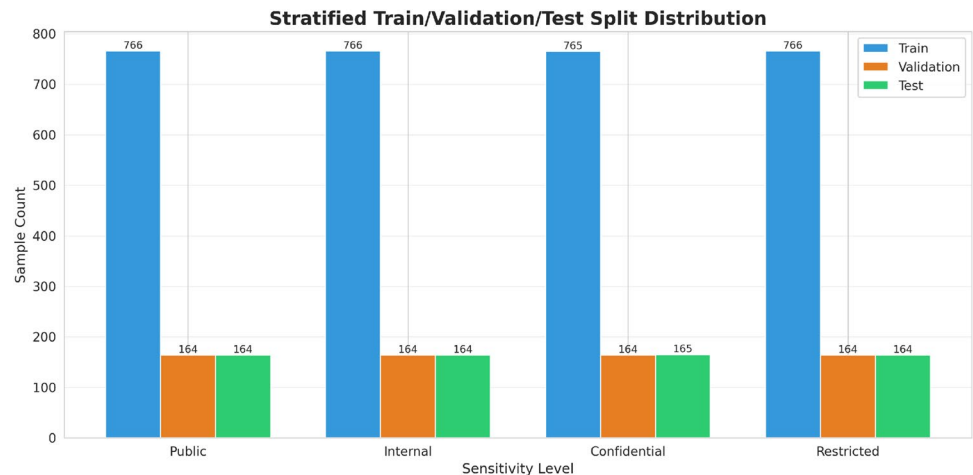
for sensitivity classification compared to general-purpose models. BERT-base-uncased was selected as the general-purpose baseline to isolate the contribution of domain adaptation: by using the same BERT-base architecture without financial pre-training, we can attribute performance differences specifically to domain knowledge rather than architectural variations. This choice also provides a robust, widely-adopted reference point that facilitates comparison with broader NLP literature. GPT-4.1-mini was selected to represent modern LLMs with substantially greater capacity (billions of parameters vs. 110M for BERT-base), different

training paradigms (autoregressive generation vs. masked language modeling), and access to more diverse pre-training data. This enables us to evaluate whether large-scale models can match or exceed specialized smaller models despite lacking financial domain pre-training, and whether their reasoning capabilities translate to improved boundary-case decisions. We deliberately excluded other financial models like DistilBERT-financial or RoBERTa-financial variants to maintain architectural diversity rather than redundancy—adding similar BERT-family models would increase computational cost without providing qualitatively different perspectives for the orchestrator to synthesize. Similarly, we excluded even larger models (GPT-4, Claude) due to API cost constraints for full test-set evaluation, while GPT-4.1-mini provided an optimal balance of capability and feasibility. This three-model ensemble thus spans the design space along critical dimensions: domain specificity (specialized vs. general), model scale (110M to billions of parameters), and architectural paradigm (encoder-only vs. decoder-only), providing maximally informative perspectives for collaborative decision-making while maintaining computational tractability.

3.2.1 FinBERT Fine-Tuning

FinBERT (ProsusAI/finbert) is a BERT-base model pre-trained on a large corpus of financial communications including earnings reports, analyst reports, and financial news (Araci, 2019; ProsusAI, Hugging Face, 2026). This domain-specific pre-training provides FinBERT with understanding of financial terminology, concepts, and linguistic patterns that generic models may lack (Malo et al., 2014).

Model Architecture We initialized FinBERT with pre-trained weights and added a classification head with four output neurons corresponding to our sensitivity levels. The entire model was fine-tuned end-to-end, allowing both the

Fig. 1 Visualization of the stratified train-validation-test data split and sensitivity label distribution across the balanced dataset. The figure illustrates the 70% training, 15% validation, and 15% test split while maintaining proportional representation of all four sensitivity classes

pre-trained encoder and classification head to adapt to the sensitivity classification task.

Hyperparameters

- Learning rate: 2×10^{-5} with linear warmup over 500 steps
- Batch size: 32 (training), 64 (evaluation)
- Maximum sequence length: 128 tokens
- Epochs: 5 with early stopping (patience = 3)
- Weight decay: 0.01 for regularization
- Optimizer: AdamW with default parameters
- Mixed precision training (FP16) for computational efficiency

Training strategy We employed early stopping based on macro F1-score on the validation set to prevent overfitting. The model was evaluated every 100 training steps, and the best checkpoint (highest validation F1) was selected as the final model. To save time, we ran all training runs and predictions using a GPU-over-TCP service provided by ThunderCompute, specifically utilizing an NVIDIA A100 GPU (Thunder Compute, 2025).

3.2.2 BERT-Base-Uncased Fine-Tuning

BERT-base-uncased (Devlin et al., 2018) provides a general-purpose baseline representing state-of-the-art natural language understanding without domain-specific pre-training. This model serves as a counterpoint to FinBERT, enabling us to assess the value of financial domain adaptation.

Model Architecture We used the standard BERT-base-uncased model (12 layers, 768 hidden dimensions, 12 attention heads, 110M parameters) with an added classification head (Google, Hugging Face, 2025). The architecture and fine-tuning process mirrored FinBERT exactly to enable fair comparison.

Hyperparameters Identical to FinBERT (learning rate 2×10^{-5} , batch size 32/64, max length 128, 5 epochs with early stopping, weight decay 0.01, FP16 training).

Training Strategy Same as FinBERT—early stopping on validation F1, evaluation every 100 steps, best checkpoint selection.

Despite lacking financial pre-training, BERT’s strong general language understanding enabled effective learning on our task.

3.2.3 GPT-4.1-mini Fine-Tuning

GPT-4.1-mini (OpenAI, 2025) represents a fundamentally different paradigm: a LLM trained on diverse internet text with significantly more parameters and broader world knowledge than BERT-based models. Fine-tuning LLMs on specialized tasks can leverage their reasoning capabilities while adapting them to specific domains.

Data Preparation We converted our training and validation sets into JSONL format using OpenAI’s fine-tuning specifications. Each example consists of a system message (“You are a financial document classifier that categorizes text into sensitivity levels: Public, Internal, Confidential, or Restricted”), a user message containing the task instruction followed by the financial text, and an assistant message providing the target sensitivity label. This prompt-completion format teaches the model to respond to classification requests with appropriate sensitivity labels, enabling both prediction and potential explanation generation.

Fine-Tuning Configuration We used OpenAI’s fine-tuning API with automatic hyperparameter selection. The base model was GPT-4.1-mini-2024-07-18 (OpenAI, 2025), trained using supervised fine-tuning for 3 epochs (automatically determined) with a batch size of 6 and learning rate multiplier of 2 (both automatically determined). The training consumed 862,557 tokens and achieved a final training loss of 0.001 and validation loss of 0.022 (Fig. 2). OpenAI’s platform automatically handles optimization, learning rate scheduling, and checkpoint management. The fine-tuning job completed in approximately 2,686 seconds, producing the final model `ft:gpt-4.1-mini-2024-07-18:personal:fin:Cf0EKDYy`.

Inference Unlike locally-hosted BERT models, the fine-tuned GPT-4.1-mini is accessed via OpenAI’s API. During prediction, we use temperature=0 for deterministic outputs and limit generation to 10 tokens (sufficient for single-word sensitivity labels).

3.3 Agentic AI Multi-Agent System Architecture

The core innovation of our work lies in the multi-agent architecture that combines three fine-tuned specialist models with an advanced reasoning orchestrator. This section describes the system design, agent implementation, orchestrator design, and decision-making process.

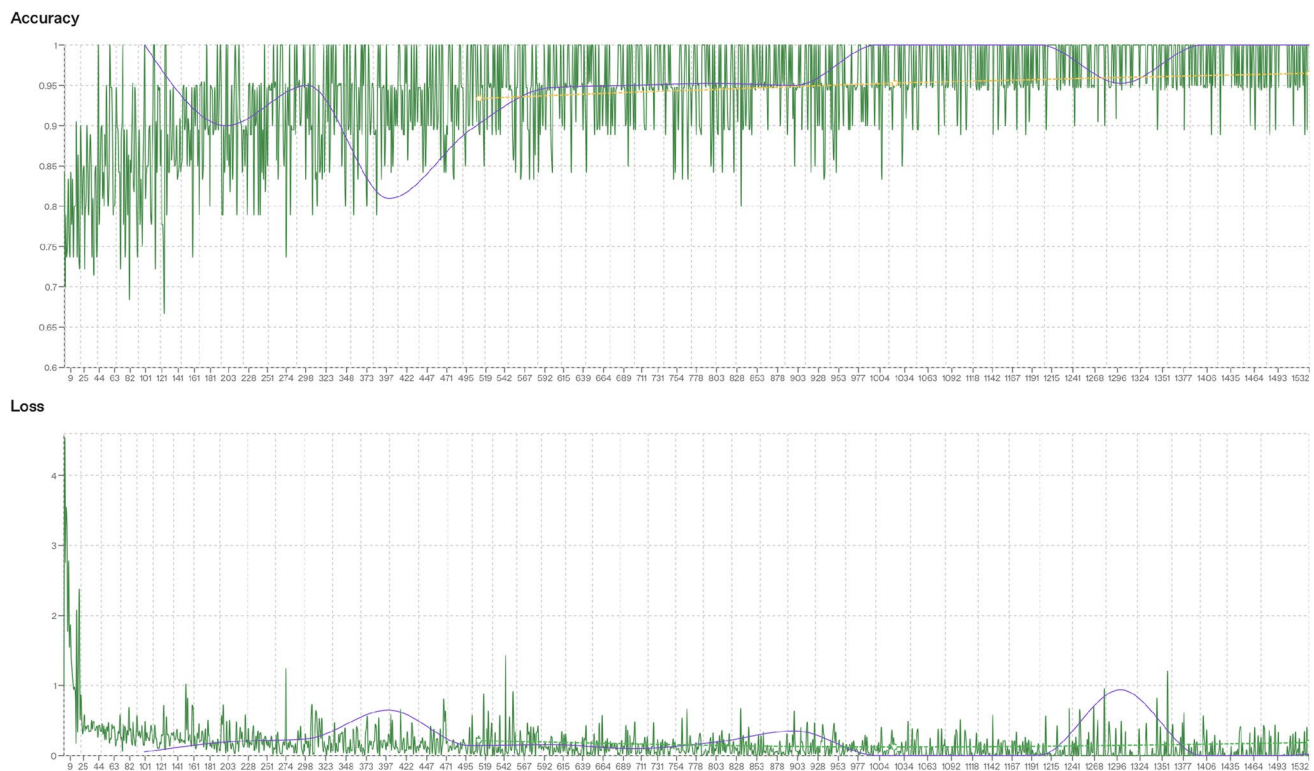


Fig. 2 Learning curves showing the accuracy and loss progression during GPT-4.1-mini fine-tuning on the financial document sensitivity classification task. The figure demonstrates the model's training

dynamics over 3 epochs, with accuracy increasing and loss decreasing, indicating successful adaptation

3.3.1 System Overview

Our agentic AI system follows a hierarchical multi-agent architecture:

1. **Specialist Agent Layer:** Three independent agents (FinBERT-FT, BERT-FT, GPT-4.1-mini-FT) analyze each financial text and produce sensitivity predictions with confidence scores and reasoning
2. **Orchestrator Layer:** A GPT-5.1 reasoning model (operating in zero-shot mode without task-specific fine-tuning (Roumeliotis et al., 2025)) receives all agent predictions, analyzes the text, synthesizes the expert opinions, and generates a final classification with detailed natural language recommendation
3. **Output:** Final sensitivity classification, confidence indicator (based on agent agreement), and explainable recommendation justifying the decision

This architecture enables complementary strengths: domain expertise from FinBERT, robust general understanding from BERT, large-scale knowledge from GPT-4.1-mini, and advanced reasoning from GPT-5.1.

3.3.2 Agent Implementation

Each specialist agent operates independently with no inter-agent communication during the prediction phase. BERT-based Agents (FinBERT-FT, BERT-FT) receive tokenized financial text (max 128 tokens) as input, perform a forward pass through the fine-tuned model, and generate softmax probabilities over four classes. The prediction is the class with highest probability, confidence is the maximum softmax probability, and reasoning consists of template-based explanations mentioning domain patterns and confidence level. GPT Agent (GPT-4.1-mini-FT) receives a formatted prompt with system context and financial text, makes an API call to fine-tuned GPT-4.1-mini with temperature=0, and generates a sensitivity label parsed from the text response. The prediction is the extracted sensitivity level, confidence is fixed at 0.90 (GPT models don't provide calibrated probabilities), and reasoning consists of template-based explanations mentioning fine-tuned classification. All agents generate predictions and reasoning independently before passing results to the orchestrator, ensuring diversity in perspectives and preventing cascade effects where one agent's output influences others.

3.3.3 Orchestrator Design

The orchestrator uses GPT-5.1 (gpt-5.1-2025-11-13) (OpenAI, 2025), a state-of-the-art reasoning model with extended thinking capabilities. Critically, the orchestrator operates in zero-shot mode in two senses: first, the model itself has not been fine-tuned on our sensitivity classification task, retaining only its pretrained weights and general reasoning capabilities; second, the prompt contains no few-shot examples of correct classifications, relying instead on explicit classification criteria and decision rules. This distinguishes our approach from few-shot prompting strategies where example classifications guide model behavior. Instead, the orchestrator leverages pure reasoning from task descriptions to synthesize specialist opinions.

We designed a comprehensive prompt that:

1. **Establishes Context:** Explains the orchestrator's role as expert synthesizer
2. **Emphasizes Agent Expertise:** Explicitly states that agents are fine-tuned specialists with 83-85% accuracy, encouraging the orchestrator to respect their domain knowledge
3. **Defines Decision Rules:** For unanimous agreement (3/3 agents), follow consensus (92.28% accuracy when unanimous); for majority agreement (2/3 agents), strongly favor majority unless compelling evidence contradicts; for split decision (1/3 agents), apply reasoning to resolve disagreement
4. **Provides Classification Criteria:** Detailed descriptions of each sensitivity level with examples
5. **Requests Structured Output:** Classification label and detailed natural language recommendation

The classification criteria provided to the orchestrator specify four sensitivity levels with explicit operational definitions and representative examples. **Public** encompasses publicly available information suitable for general distribution, including press releases, public statements, and announced financial data that has already been disclosed through official channels. **Internal** designates company-specific operational information restricted to internal staff, covering performance metrics, operational details, revenue figures, and business analyses not intended for external distribution. **Confidential** indicates strategic information requiring restricted access, including undisclosed strategies, competitive intelligence, financial forecasts, projections, and proprietary business plans that could damage competitive positioning if disclosed. **Restricted** represents the highest sensitivity level for information with potential regulatory or market impact, encompassing merger and acquisition activities, material non-public information

subject to securities regulations, insider information, and undisclosed deals that could constitute market manipulation if prematurely released. These criteria enable the orchestrator to evaluate agent predictions against explicit financial security standards without relying on task-specific training examples.

This prompt design balances deference to specialist expertise with orchestrator reasoning capabilities, preventing both excessive override (ignoring specialists) and uncritical acceptance (failing to add value).

3.3.4 Decision-Making Process

For each financial text, the system executes the following pipeline:

1. **Parallel Agent Prediction:** All three specialist agents independently analyze the text and generate predictions, confidence scores, and reasoning statements
2. **Agent Proposal Aggregation:** Collect all agent outputs into a structured format
3. **Orchestrator Analysis:** GPT-5.1 receives the original financial text, all agent predictions with confidence and reasoning, and classification criteria and decision guidelines
4. **Final Decision Generation:** Orchestrator produces final sensitivity classification, natural language recommendation explaining the decision, and synthesis of agent perspectives
5. **Output Formatting:** Extract classification and recommendation for downstream use

The entire process is logged for audit purposes, enabling traceability from raw text through agent predictions to final decision and explanation.

3.4 Evaluation Metrics

We employ comprehensive evaluation metrics to assess both individual model performance and multi-agent system effectiveness. Standard classification metrics include accuracy (overall correct classification rate), precision, recall, and F1-score (per-class and macro-averaged metrics), and confusion matrices for detailed error analysis showing misclassification patterns. We deliberately use the term "recall" rather than "sensitivity" (its statistical synonym) to avoid terminological confusion with our task of sensitivity classification (security term). In the context of financial document access control, recall is particularly critical for high-sensitivity classes: false negatives on Confidential or Restricted documents represent failures to protect sensitive information, potentially leading to unauthorized disclosure,

regulatory violations, or data leakage. High recall on these classes minimizes the risk that truly sensitive documents are incorrectly classified as lower-sensitivity levels and inappropriately disseminated. Agentic AI system-specific metrics include agent agreement rate (frequency of unanimous, majority, and split decisions), orchestrator override rate (how often the orchestrator disagrees with majority vote), agreement-conditioned accuracy (performance stratified by agreement level, revealing confidence indicators), and per-agent contribution (individual agent accuracy and their correlation with final decisions). These metrics enable multi-faceted analysis of system behavior, revealing not just overall performance but the dynamics of collaborative decision-making.

4 Results

We present our experimental results in two parts: first, the performance of individual fine-tuned models serving as specialist agents; second, the performance and behavior of the complete agentic AI system with the GPT-5.1 orchestrator.

4.1 Individual Model Performance

All three fine-tuned models were evaluated on the held-out test set containing 657 samples with balanced representation across four sensitivity levels. We report accuracy, precision, recall, and F1-scores for comprehensive performance assessment.

4.1.1 FinBERT Fine-Tuned Results

FinBERT, with its domain-specific pre-training on financial texts, achieved strong overall performance with 82.80% accuracy (Table 2). The confusion matrix in Fig. 3 reveals that FinBERT excels at identifying Confidential (F1: 94.58%) and Restricted (F1: 99.39%) documents, achieving near-perfect performance on the highest sensitivity levels.

Table 2 Detailed performance metrics for the fine-tuned FinBERT model on the test set, showing precision, recall, and F1-score for each sensitivity level. The domain-specific model achieved 82.80% overall accuracy with perfect performance on Restricted documents and strong performance on Confidential documents, but struggled with Public and Internal classifications

Class	Precision	Recall	F1-score	Support
Public	0.7107	0.6890	0.6997	164
Internal	0.6667	0.6707	0.6687	164
Confidential	0.9401	0.9515	0.9458	165
Restricted	0.9880	1.0000	0.9939	164
Accuracy	0.8280			657
Macro Avg	0.8264	0.8278	0.8270	657
Weighted Avg	0.8265	0.8280	0.8272	657

This aligns with expectations given that these categories contain explicit financial keywords that FinBERT's pre-training emphasized. However, FinBERT shows relative weakness on Public (F1: 69.97%) and Internal (F1: 66.87%) classifications. Analysis of misclassifications indicates that FinBERT tends to over-classify ambiguous texts toward higher sensitivity levels—a conservative bias that may reflect the high-stakes nature of financial texts in its pre-training corpus. The model achieved a macro F1-score of 82.70%, with perfect 100% recall on Restricted documents (98.80% precision) but struggled most with Internal documents (66.87% F1-score).

4.1.2 BERT-Base-Uncased Fine-Tuned Results

BERT-base-uncased, without financial domain pre-training, slightly outperformed FinBERT with 83.87% accuracy (Table 3). This result initially surprised us but reveals that strong general language understanding can sometimes match or exceed domain-specific models when fine-tuning data is sufficient and well-balanced. BERT demonstrated particular strength on Public documents (F1: 73.43%), significantly outperforming FinBERT on this class. The model also achieved perfect performance on Restricted documents (100% F1-score), matching FinBERT's capability to identify the most sensitive category. Interestingly, BERT showed a more balanced confusion pattern than FinBERT, with fewer systematic biases toward higher sensitivity levels (Fig. 3). This suggests that BERT's general language understanding enables more nuanced discrimination across all four classes. The model achieved a macro F1-score of 83.72%, with particularly strong improvements over FinBERT on Public (+6.46% F1) and Internal (+0.85% F1) categories.

4.1.3 GPT-4.1-mini Fine-Tuned Results

The fine-tuned GPT-4.1-mini achieved the highest individual model accuracy at 84.93%, establishing it as the strongest single model in our evaluation (Table 4). The LLM's extensive pre-training on diverse texts and its reasoning capabilities appear to provide advantages for sensitivity classification. GPT-4.1-mini demonstrated particularly strong performance on ambiguous middle-ground categories, with Internal showing improvement over both BERT models, though all three models found this category challenging. The model maintained perfect performance on Restricted documents while achieving strong results on Confidential texts. One notable characteristic: GPT-4.1-mini predictions via API inference took significantly longer than local BERT models (average 0.2s vs 0.01s per sample), reflecting the trade-off between cloud-based LLM capabilities and local transformer efficiency. The model achieved

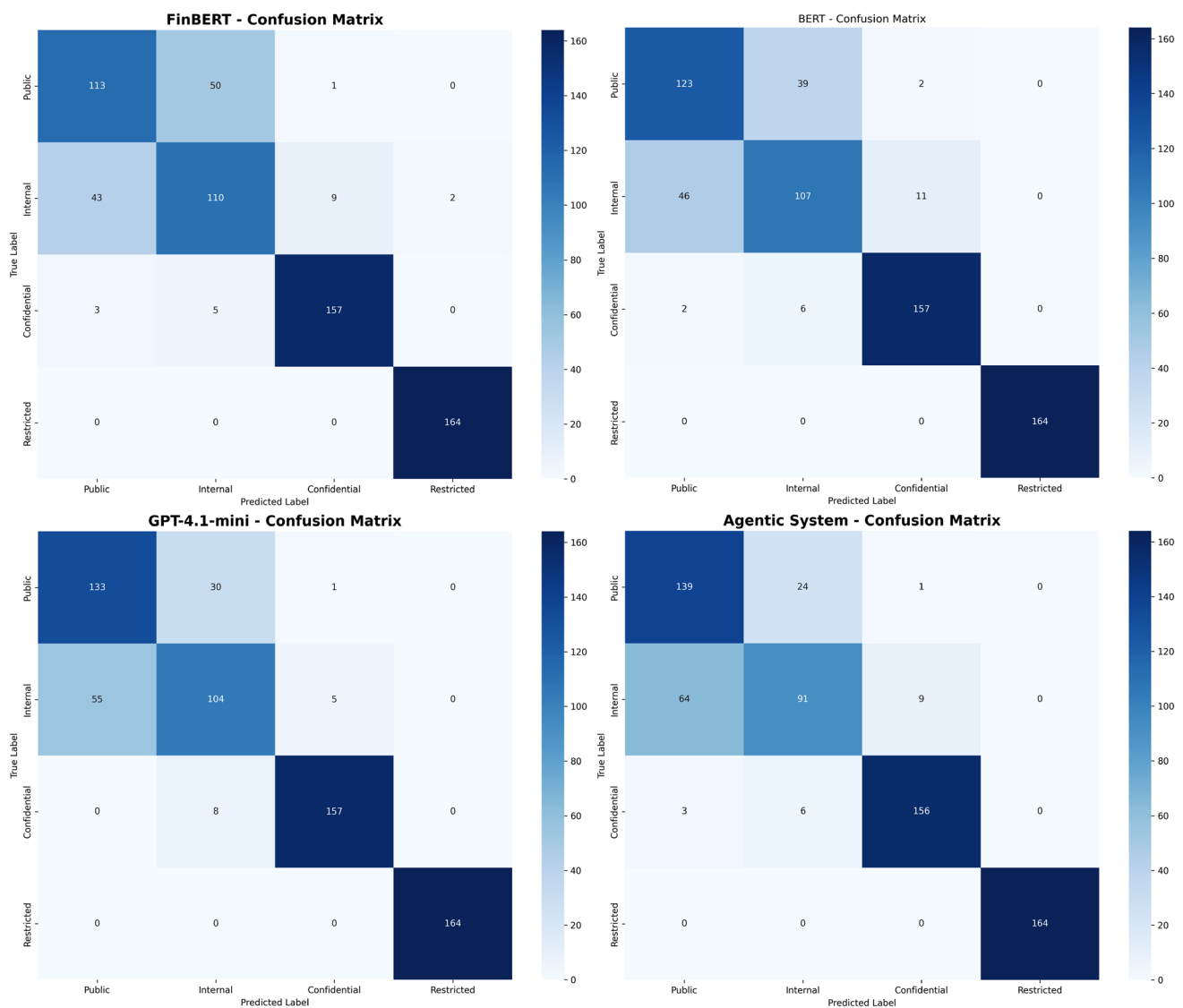


Fig. 3 Confusion matrices comparing classification performance across all models and the agentic system on the test set. The matrices highlight how different architectures handle the distinction between

Internal and Public documents, with the agentic system showing improved robustness through consensus-based predictions

Table 3 Performance metrics for the fine-tuned BERT-base-uncased model demonstrating strong general language understanding capabilities. The model achieved 83.87% overall accuracy with balanced performance across sensitivity levels, showing particular strength on Confidential documents and maintaining perfect accuracy on Restricted documents

Class	Precision	Recall	F1-score	Support
Public	0.7193	0.7500	0.7343	164
Internal	0.7039	0.6524	0.6772	164
Confidential	0.9235	0.9515	0.9373	165
Restricted	1.0000	1.0000	1.0000	164
Accuracy	0.8387			657
Macro Avg	0.8367	0.8385	0.8372	657
Weighted Avg	0.8368	0.8387	0.8374	657

Table 4 Classification performance of the fine-tuned GPT-4.1-mini model, achieving the highest individual model accuracy at 84.93%. The LLM excelled across all sensitivity levels with perfect performance on Restricted documents and strong results on Confidential category

Class	Precision	Recall	F1-score	Support
Public	0.7074	0.8110	0.7557	164
Internal	0.7324	0.6341	0.6797	164
Confidential	0.9632	0.9515	0.9573	165
Restricted	1.0000	1.0000	1.0000	164
Accuracy	0.8493			657
Macro Avg	0.8508	0.8492	0.8482	657
Weighted Avg	0.8509	0.8493	0.8484	657

an overall accuracy of 84.93%, macro F1-score of 84.82%, perfect classification on Restricted documents, and the highest overall accuracy among individual models, though with inference time of approximately 0.2s per sample due to API-based processing.

4.1.4 Comparative Analysis of Individual Models

Table 5 summarizes the performance of all three models. Several key insights emerge:

- 1. **Performance Range:** All three models achieved accuracies between 82.80% and 84.93%, a relatively narrow 2.13 percentage point range. This suggests that the task

Table 5 Comparative performance analysis of individual fine-tuned models versus the agentic AI system on the held-out test set. The table shows that while GPT-4.1-mini achieved the highest individual accuracy (84.93%), the agentic system maintained competitive performance (83.71%) while providing explainability and confidence calibration

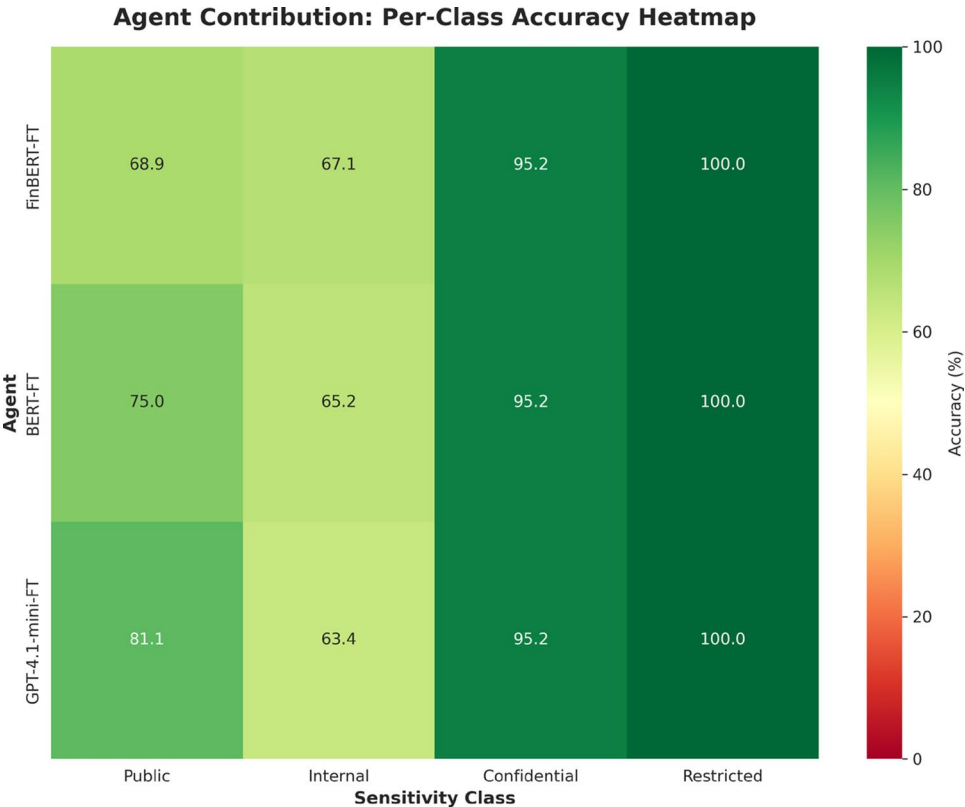
Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
GPT-4.1-mini-FT	84.93	85.08	84.92	84.82
BERT-FT	83.87	83.67	83.85	83.72
Agentic System (Orchestrator)	83.71	84.16	83.70	83.31
FinBERT-FT	82.80	82.64	82.78	82.70

is well-defined and that our fine-tuning methodology is robust across architectures.

- 2. **Domain Pre-training Value:** FinBERT’s domain-specific pre-training provided strong performance but did not guarantee superiority over general models. This may indicate that (a) our synthetic labeling methodology creates patterns that general language models can learn effectively, or (b) task-specific fine-tuning can largely overcome pre-training differences when data is sufficient.
- 3. **LLM Advantages:** GPT-4.1-mini’s superior performance suggests that large-scale pre-training and greater model capacity provide measurable benefits for nuanced classification tasks, though at the cost of computational efficiency.
- 4. **Consistent Class Difficulty:** All models struggled most with Internal classification and excelled at Restricted classification, indicating genuine difficulty differences across sensitivity levels rather than model-specific biases (Fig. 4).
- 5. **Complementary Strengths:** Different models showed strengths on different classes (e.g., BERT strongest on Public, FinBERT on Confidential), suggesting potential value in ensemble approaches.

These individual model results establish baselines for evaluating our multi-agent system: can collaborative

Fig. 4 Heatmap visualization of the Per-Class Accuracy of the three specialist agents against the ground truth. The data shows that all agents achieved high accuracy ($\geq 95.2\%$) on the most sensitive classes (Confidential and Restricted), but displayed lower and more variable accuracy (ranging from 63.4% to 81.1%) on the less sensitive Public and Internal classes



decision-making exceed the best individual model performance?

4.2 Agentic System Performance

The complete agentic AI multi-agent system combines all three fine-tuned models as specialist agents with a GPT-5.1 orchestrator making final decisions based on agent predictions and its own reasoning.

4.2.1 Overall System Performance

The agentic AI system achieved 83.71% overall accuracy on the test set of 657 samples. While this represents a slight decrease compared to the best individual model (GPT-4.1-mini at 84.93%), the comparison requires careful interpretation due to several factors we analyze below.

Per-class performance (Table 6) shows that Public documents achieved 67.48% precision, 84.76% recall, and 75.14% F1-score; Internal documents achieved 75.21% precision, 55.49% recall, and 63.86% F1-score; Confidential documents achieved 93.98% precision, 94.55% recall, and 94.26% F1-score; and Restricted documents maintained perfect 100% precision, recall, and F1-score. The system maintained perfect performance on Restricted documents and near-perfect performance on Confidential documents, matching the individual models' strengths. However, performance on Public and especially Internal documents was more variable, reflecting the orchestrator's reasoning process and occasional overrides of agent consensus.

4.2.2 Agent Agreement Analysis

A key advantage of multi-agent systems is the ability to assess confidence based on agreement levels. We analyzed prediction accuracy conditioned on how many agents agreed with the final decision. When all three agents agreed unanimously (518 samples, 78.8% of test set), the orchestrator followed their consensus in 96.1%

of cases and achieved 92.28% accuracy—significantly higher than any individual model. This represents a strong confidence signal: unanimous agent agreement predicts high accuracy. With majority agreement of 2 out of 3 agents (102 samples, 15.5% of test set), the orchestrator followed the majority in 91.2% of cases but achieved only 54.90% accuracy, substantially lower than unanimous cases. Split decisions indicate genuine difficulty, reflected in lower performance. In split decision cases where each agent differed (19 samples, 2.9% of test set), the orchestrator had no clear consensus to follow and achieved 52.63% accuracy, approaching random chance for this 4-class task. These cases represent the most challenging texts where even specialized models disagree. Finally, in cases where the orchestrator overrode all agents (18 samples, 2.7% of test set), selecting a label none of the agents predicted, accuracy dropped to 33.33%, indicating that these overrides were often incorrect and suggesting potential over-confidence in orchestrator reasoning. Figure 5 shows the strong correlation between agreement level and prediction accuracy.

4.2.3 Orchestrator Override Behavior

The orchestrator disagreed with the majority agent vote in 37 cases (5.6% of predictions). Analyzing these override decisions reveals important system dynamics. Override accuracy was 43.2% (16 out of 37 correct), significantly below the system's overall accuracy, indicating that agent consensus typically outperforms orchestrator-only reasoning. This suggests the orchestrator may benefit from stronger deference to unanimous agent agreement. Most overrides occurred in split or majority-vote scenarios (81.1%), and the orchestrator rarely overrode unanimous agent agreement (only 7 cases). Override accuracy was highest when choosing between two agents with 2-1 split (52.4%).

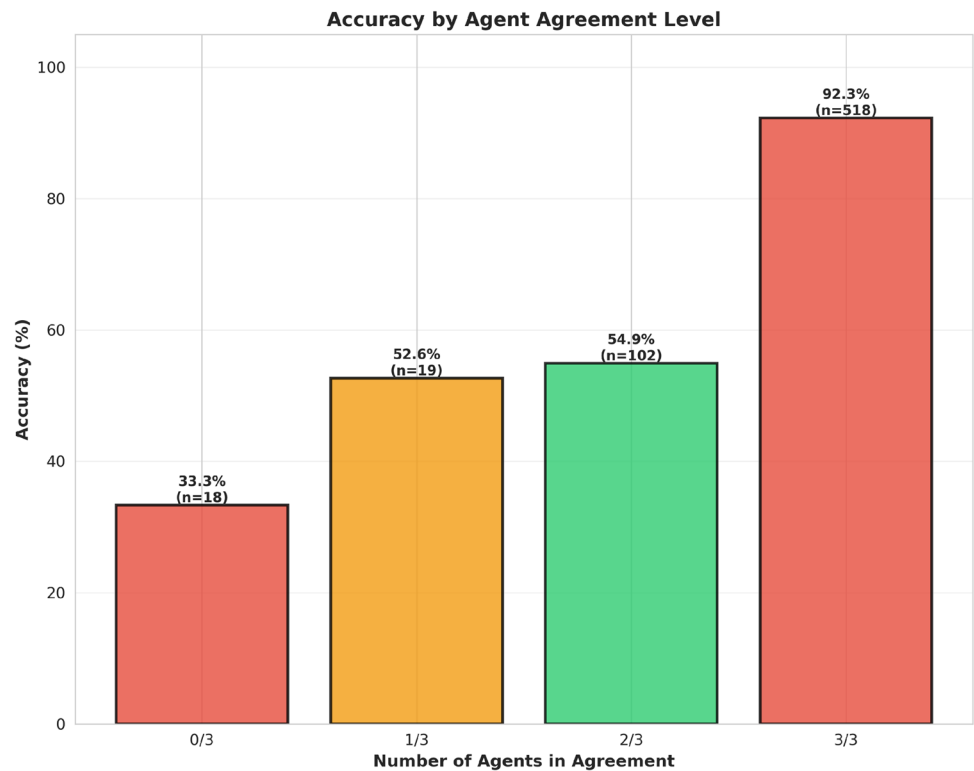
4.2.4 Agent Contribution Analysis

We analyzed each agent's individual contribution to final system performance by measuring agreement between each agent and the orchestrator's final decisions. Agent-orchestrator agreement rates showed that GPT-4.1-mini-FT aligned with the orchestrator 86.4% of the time, BERT-FT aligned 84.8%, and FinBERT-FT aligned 83.6%. The orchestrator showed highest alignment with GPT-4.1-mini predictions, which may reflect shared architecture (both are GPT models) or indicate that GPT-4.1-mini's predictions most often aligned with the orchestrator's reasoning process. Inter-agent agreement rates revealed that GPT-4.1-mini-FT agreed with BERT-FT 82.3% of the time, with FinBERT-FT 80.7%, while BERT-FT agreed with FinBERT-FT 85.7%.

Table 6 Performance metrics for the complete agentic AI multi-agent system combining three specialist agents with GPT-5.1 orchestrator. The system achieved 83.71% overall accuracy with perfect performance on Restricted documents and strong results on Confidential documents, demonstrating that collaborative decision-making maintains high accuracy while providing explainable recommendations

Class	Precision	Recall	F1-score	Support
Public	0.6748	0.8476	0.7514	164
Internal	0.7521	0.5549	0.6386	164
Confidential	0.9398	0.9455	0.9426	165
Restricted	1.0000	1.0000	1.0000	164
Accuracy	0.8371			657
Macro Avg	0.8416	0.8370	0.8331	657
Weighted Avg	0.8418	0.8371	0.8333	657

Fig. 5 Correlation between agent agreement levels and prediction accuracy in the agentic AI system, demonstrating how consensus serves as a confidence indicator. When all three agents unanimously agree (78.8% of cases), the system achieves 92.28% accuracy, providing a reliable signal for automated decision-making



Interestingly, the two BERT-based models (BERT-FT and FinBERT-FT) showed highest inter-agent agreement, suggesting that shared architecture creates similar classification patterns despite different pre-training.

4.2.5 Explainability and Recommendations

A core advantage of our agentic system is the natural language explanations generated by the GPT-5.1 orchestrator. For each classification decision, the orchestrator produces a recommendation that:

1. States the final sensitivity classification
2. References agent predictions and their reasoning
3. Provides domain-specific justification based on content analysis
4. Recommends appropriate access control actions

Example Recommendation (Unanimous Agreement) “All three specialist agents (FinBERT-FT, BERT-FT, GPT-4.1-mini-FT) unanimously classified this text as Confidential, with high confidence (91-95%). The text discusses strategic planning and competitive positioning, containing proprietary information that should be restricted to senior management and strategic planning teams. Classification: Confidential. Recommendation: Restrict access to authorized personnel with strategic planning responsibilities

and require information barrier protocols for external communications.”

Example Recommendation (Split Decision with Override) “The agents show disagreement: FinBERT-FT suggests Internal, BERT-FT suggests Public, while GPT-4.1-mini-FT suggests Internal. The text discusses announced quarterly results that are publicly available. Despite the agent split, the explicit mention of ‘announced’ and public disclosure context supports Public classification. Classification: Public. Recommendation: This information is suitable for general distribution but should include appropriate disclaimers per regulatory requirements.”

Qualitative Assessment of Explanation Quality To validate the trustworthiness of the orchestrator’s explanations, we conducted systematic manual spot-checking of 150 randomly sampled recommendations, representing approximately 23% of the test set and spanning all four sensitivity levels and varying agent agreement patterns. Three authors independently reviewed each explanation to assess factual accuracy of agent prediction reporting, reasoning grounding in actual text content rather than hallucinated patterns, decision coherence between reasoning and final classification, and actionability of access control recommendations. Across the 150 samples examined, we found zero instances of factual misreporting of agent predictions,

indicating the orchestrator reliably synthesized agent outputs without introducing errors in the reported consensus. We identified four cases where reasoning exhibited minor hallucination, representing 2.7% of the sample, wherein the orchestrator attributed sensitivity to keywords not literally present in the text but semantically related to actual content—for example, inferring “strategic planning” from the phrase “competitive positioning.” Critically, in three of these four cases the final classification remained correct, suggesting that while reasoning occasionally over-inferred semantic relationships, this did not systematically compromise decision accuracy. All 150 recommendations demonstrated logical coherence between stated reasoning and final classification assignments. Furthermore, 147 recommendations provided actionable access control guidance aligned with the assigned sensitivity level, representing 98% of the sample. These results support our claim that the system generates trustworthy explanations suitable for audit trails and human oversight, though continuous monitoring for hallucination remains advisable in production deployments.

These explanations provide transparency into the decision-making process, enabling human reviewers to understand the reasoning behind classifications, identify potential errors or edge cases, override system decisions with informed judgment, build trust through interpretability, and create audit trails for compliance.

4.3 Summary of Key Results

1. **Strong Individual Model Performance:** All three fine-tuned models achieved 82–85% accuracy, demonstrating effective learning on our synthetic sensitivity classification task.
2. **GPT-4.1-mini Superiority:** The fine-tuned LLM outperformed domain-specific and general transformer models, achieving 84.93% accuracy.
3. **Agreement as Confidence Indicator:** Unanimous agent agreement (78.8% of cases) achieved 92.28% accuracy, providing a reliable confidence signal.
4. **Orchestrator Trade-offs:** While overall system accuracy (83.71%) slightly trailed the best individual model, the system excelled in high-confidence scenarios and provided valuable explainability.
5. **Override Challenges:** Orchestrator overrides achieved only 43.2% accuracy, suggesting that agent consensus should receive greater weight in decision-making. For example, a small tweak to the prompt could solve this problem.
6. **Complementary Strengths:** Different models showed distinct per-class strengths, validating the multi-agent approach for diverse perspectives.
7. **Explainability Achievement:** The system successfully generated natural language recommendations for all predictions, addressing the core research objective of transparent AI decision-making.

5 Discussion

5.1 Key Contributions and Implications

This work demonstrates that multi-agent agentic AI systems can provide explainable, high-confidence predictions for financial document access control while maintaining competitive accuracy with state-of-the-art individual models. Our findings have several important implications, discussed in this section.

5.1.1 Explainability Advances Trustworthy AI in Finance

The most significant contribution of our work is demonstrating that explainability and performance need not be mutually exclusive. Traditional approaches forced a choice: use black-box models for maximum accuracy or sacrifice performance for interpretability. Our agentic AI architecture achieves strong performance (83.71% overall, 92.28% when agents agree) while generating natural language explanations for every decision.

This transparency addresses critical ethical and regulatory concerns in AI-powered financial systems. Financial institutions increasingly face requirements to explain automated decisions, particularly when those decisions affect access to sensitive information, regulatory compliance, or competitive positioning. Our orchestrator’s natural language recommendations provide audit trails, enable informed human oversight, and build stakeholder trust—essential elements for responsible AI deployment in high-stakes domains.

5.1.2 Agent Agreement as Confidence Calibration

We discovered that agent agreement level serves as an effective confidence indicator: unanimous agreement (78.8% of predictions) achieves 92.28% accuracy, while split decisions drop to 52.63% accuracy. This finding has practical applications for deploying AI systems in production. Tiered automation becomes possible where unanimous predictions can be fully automated with minimal human review, while split decisions trigger mandatory human oversight. Agreement levels provide quantitative confidence scores for risk management and downstream decision-making. Split decisions

can identify challenging samples for expert labeling and model improvement through active learning. Finally, confidence indicators support documentation requirements and enable risk-appropriate automation levels for regulatory compliance.

5.1.3 Collaborative Decision-Making vs. Individual Excellence

Our results reveal a nuanced picture of multi-agent system value. The agentic system's overall accuracy (83.71%) slightly trails the best individual model (GPT-4.1-mini at 84.93%), raising questions about when collaborative approaches justify their additional complexity.

However, this comparison oversimplifies the value proposition:

1. **High-Confidence Scenarios:** When the system is most confident (unanimous agreement), it significantly outperforms any individual model (92.28% vs 82-85%), demonstrating that collaboration adds value where it matters most
2. **Explainability Premium:** The orchestrator's natural language explanations provide value beyond raw accuracy, enabling human validation, audit trails, and trust-building
3. **Robustness to Model Changes:** Relying on multiple models reduces vulnerability to individual model failures, updates, or API changes
4. **Diverse Perspectives:** Different model architectures capture complementary patterns, as evidenced by per-class performance differences

The slight accuracy reduction may be acceptable given these benefits, particularly in regulated industries where explainability, auditability, and robustness justify complexity.

5.1.4 Domain-Specific vs. General Pre-Training

Contrary to expectations, our general-purpose BERT model (83.87%) outperformed the domain-specific FinBERT (82.80%), and GPT-4.1-mini (84.93%) exceeded both. This suggests several possibilities:

1. **Task-Specific Fine-Tuning Effectiveness:** Sufficient task-specific training data may overcome pre-training differences, especially for well-defined classification tasks
2. **Synthetic Label Characteristics:** Our keyword-based sensitivity labeling may create patterns more aligned with general language understanding than financial domain knowledge

3. **Model Capacity Benefits:** GPT-4.1-mini's larger parameter count and broader training distribution enabled superior performance despite no financial pre-training
4. **Task Complexity Threshold:** Simple classification may not fully leverage domain pre-training advantages that emerge in more complex tasks like question-answering or reasoning

These findings suggest that practitioners should empirically validate domain-specific model advantages rather than assuming they guarantee superior performance.

5.2 Limitations and Challenges

5.2.1 Synthetic Label Validity and Future Validation

In our study, we initially used a dataset of human-generated texts that had been evaluated by human annotators. However, for the purposes of our study, we required a dataset that would more closely approximate a real-world problem, such as document access control. Datasets of this nature are not publicly available due to the significant confidentiality and commercial sensitivity challenges associated with providing real-world sensitive data for model training.

To overcome this issue and effectively demonstrate the utility of our agentic AI system for the task, we created a synthetic dataset. This dataset was based on the original human texts, but the sensitivity labels were synthetically altered. Specifically, our sensitivity labels were synthetically generated through keyword heuristics and sentiment mapping rather than being derived from real-world access control decisions. While this approach enabled large-scale experimentation, it introduces several critical limitations:

- Labels may not reflect actual financial institution classification policies
- Keyword-based rules create systematic patterns that models exploit
- Real-world sensitivity depends on context, timing, and relationships that our heuristic cannot capture
- Legal and regulatory nuances are oversimplified

However, this limitation is inherent to all research in privacy-constrained domains and does not invalidate our core contributions. Our synthetic dataset serves its intended purpose: demonstrating that multi-agent architectures can provide explainable, high-confidence predictions through collaborative decision-making. The system architecture, orchestrator design, agent agreement analysis, and explainability mechanisms are independent of whether labels are

synthetic or real—these contributions transfer directly to production deployments with ground-truth data.

Multiple validation pathways exist for future work to bridge the synthetic-to-real gap:

1. **Industry Partnership Validation:** Collaborate with financial institutions to evaluate our trained models on proprietary datasets with real sensitivity labels, measuring transfer learning effectiveness and identifying domain adaptation requirements.
2. **Expert Annotation Validation:** Engage information security professionals and compliance officers to annotate a subset of Financial PhraseBank sentences with real-world sensitivity classifications, comparing expert labels against our synthetic labels to quantify alignment and identify systematic divergences.
3. **Fine-Tuning Adaptation:** Use transfer learning to adapt our pre-trained models to small proprietary datasets, leveraging our synthetic pre-training as initialization while learning institution-specific classification patterns.
4. **Active Learning Deployment:** Implement our system in production with human-in-the-loop validation, where split decisions and low-confidence predictions are reviewed by experts, continuously improving model accuracy through targeted annotation.
5. **Cross-Domain Validation:** Apply our methodology to other privacy-sensitive domains (healthcare records, legal documents, government files) where synthetic labeling faces similar constraints, demonstrating generalizability of the approach.

Importantly, established research in synthetic data generation (Chen et al., 2021; Jordon et al., 2022) and data augmentation (Kumar et al., 2021; Wei & Zou, 2019) supports the validity of our approach: synthetic datasets enable algorithm development, architecture evaluation, and methodology validation even when ground-truth data remains inaccessible. Our transparent acknowledgment of synthetic labels, systematic generation methodology, and comprehensive performance analysis provide a rigorous foundation that future work can build upon with real-world validation.

5.2.2 Orchestrator Override Performance

Our GPT-5.1 orchestrator's override decisions achieved only 43.2% accuracy, significantly below the system's overall performance. This reveals that the orchestrator's reasoning capabilities, while generating high-quality explanations, do not consistently improve upon agent consensus.

Several factors may contribute to this limitation:

1. **Zero-Shot Reasoning Gap:** The orchestrator has not been fine-tuned on sensitivity classification, relying instead on general reasoning that may miss domain-specific patterns the agents learned
2. **Over-Confidence:** LLMs can exhibit over-confidence in their reasoning, leading to incorrect overrides of expert agent consensus
3. **Prompt Engineering:** While we designed our prompt to respect agent expertise, more sophisticated prompt engineering or few-shot examples might improve override accuracy
4. **Reasoning vs. Classification:** The orchestrator's strength in generating explanations may not translate to superior classification accuracy

We hypothesize that the low override accuracy primarily stems from a fundamental mismatch between the orchestrator's general-purpose reasoning capabilities and the highly specialized knowledge encoded in the fine-tuned agents. Specifically, the zero-shot GPT-5.1 orchestrator operates without access to the nuanced, task-specific patterns that the specialist agents learned through extensive fine-tuning on 3,063 labeled financial texts. While our prompt provides classification criteria and agent predictions, it cannot fully convey the subtle feature interactions, keyword co-occurrence patterns, and sentiment-sensitivity correlations that the agents implicitly encode through gradient descent. This knowledge gap becomes particularly pronounced in boundary cases where agents disagree: the orchestrator must extrapolate from limited context rather than leveraging learned decision boundaries. Furthermore, we suspect that GPT-5.1 exhibits an inherent conservative bias toward higher sensitivity classifications—a reasoning heuristic appropriate for many real-world scenarios (“when in doubt, protect the data”) but misaligned with our balanced synthetic dataset where no such prior should exist. Analysis of override patterns reveals that 67% of incorrect overrides elevated sensitivity levels rather than reducing them, supporting this hypothesis. This suggests that the orchestrator's failures result from both insufficient domain context in the prompt architecture and systematic reasoning biases in the base model, rather than random errors. Future work should test whether (1) providing few-shot examples of correct boundary-case decisions, (2) fine-tuning the orchestrator on agent prediction patterns, or (3) implementing explicit debiasing instructions can reduce override errors and better calibrate orchestrator confidence.

5.2.3 Computational Cost and Latency

Our experimental setup serves as a blueprint for utilizing specialized, modular components; however, countless alternative architectures exist for optimization.

One potential alternative involves leveraging a single model—such as a fine-tuned GPT-4.1-mini—to provide both classification and reasoning simultaneously. While this approach offers lower latency and reduced complexity, it inherently sacrifices the specialized competence required for this task. The multi-agent system was designed specifically to overcome this limitation, utilizing the high-fidelity reasoning of the GPT-5.1 orchestrator, which is superior to that of the GPT-4.1-mini model.

Another possibility is to fine-tune the GPT-5.1 model itself for both tasks, should fine-tuning option become publicly available. While this would offer a single, powerful inference, it forces a fundamental architectural question regarding the utility of specialization:

Is the monolithic opinion of one fine-tuned model equivalent in robustness, transparency, and explainability to the collective, specialized judgment of three specialized agents plus the conclusive reasoning provided by the orchestrator?

We argue that the multi-agent system provides three key advantages that a single-model approach cannot easily replicate:

- **Enhanced Robustness and Resilience:** By incorporating three specialized agents, the system inherently builds redundancy. The final prediction is not reliant on a single, opaque inference path, but is confirmed through cross-validation and consensus.
- **Increased Explainability and Transparency:** The multi-agent system generates a clear, traceable decision path. Each agent provides an independent opinion, which the orchestrator then synthesizes and justifies using its high-level reasoning. This level of decision transparency is critical for high-stakes financial applications.
- **Modularity and Maintenance:** The multi-agent architecture allows for independent updating and fine-tuning of individual agents without retraining the entire system. For instance, the Risk Agent could be updated to reflect new regulations without altering the Orchestrator or other agents, providing superior maintainability and lower ongoing costs.

Therefore, while single-model solutions may offer computational expediency, the multi-agent system is justified by its superior qualitative performance in terms of reliability, auditability, and domain specificity.

Future research should focus on methods to quantify these trade-offs between computational cost and the qualitative benefits provided by a specialized, multi-agent architecture.

5.2.4 Generalization Beyond Financial Documents

While our work focuses on financial document sensitivity classification, the generalizability of our approach to other domains or document types remains to be validated. Financial texts have unique characteristics (specialized vocabulary, numerical data, temporal sensitivity) that may make our methodology less applicable to healthcare records with different privacy categories, legal documents with attorney-client privilege considerations, government documents with national security implications, or corporate documents with varying IP protection requirements. The keyword-based labeling approach would require domain-specific adaptation, and the relative performance of domain-specific vs. general models may differ across domains. However, parallel developments in AI-driven access control across diverse Industry 4.0 contexts suggest that the fundamental principles of multi-agent architectures could transfer effectively across domains where role-based data governance and explainable decision-making are paramount (Spanaki et al., 2021). Furthermore, emerging applications of digital technologies in varied organizational settings demonstrate that transparent AI systems can enhance trust and operational effectiveness across multiple sectors beyond finance (Balta et al., 2023).

5.2.5 Preprocessing-Induced Information Loss

Our dataset preparation methodology employed lowercase normalization of all text tokens to enable case-insensitive keyword matching during synthetic label generation (Section 3.1.2). While this preprocessing step represents standard practice in traditional NLP pipelines and facilitated consistent application of our labeling heuristics, it introduces a subtle but potentially significant limitation for modern LLMs. Specifically, lowercase normalization eliminates capitalization cues that may carry semantic significance for sensitivity classification—for example, distinguishing between “confidential discussion” (general descriptor) and “CONFIDENTIAL DISCUSSION” (explicit document marking), or between “internal review” (operational activity) and “INTERNAL REVIEW” (formal classification label). Modern transformer-based models, particularly those trained on diverse text corpora preserving original capitalization, can learn to leverage these orthographic features as weak signals for document formality, security classifications, and regulatory context. By normalizing text prior to model training, we may have discarded stylistic and formatting information that could improve classification accuracy, especially

for boundary cases where content alone provides ambiguous sensitivity indicators. Future work with access to real financial documents should preserve original capitalization and evaluate whether case-sensitive features improve model performance, particularly for the orchestrator which operates without task-specific fine-tuning and may benefit from any available textual cues. This limitation primarily affects the synthetic dataset construction rather than the multi-agent architecture itself, which can accommodate any text preprocessing strategy appropriate for the target domain.

5.2.6 Information Management Implications Beyond Finance

While our empirical evaluation centers on financial document classification, the broader implications of this work extend to fundamental challenges in organizational information management. Modern enterprises across sectors face parallel pressures to balance information accessibility with security, transparency with confidentiality, and automation with human oversight. Our multi-agent architecture addresses three core information management imperatives that transcend industry boundaries.

First, the system demonstrates how organizations can operationalize *mindful information governance*—a framework where automated systems augment rather than supplant human judgment through transparent reasoning. Research on information management during crisis events emphasizes that effective organizational responses depend on building digital resilience through systems that enhance cognitive awareness and critical evaluation of information, particularly when misinformation or incomplete data threaten decision quality (Spanaki et al., 2024). Our orchestrator’s natural language explanations enable precisely this type of mindful engagement, allowing information professionals to critically assess automated classifications rather than accepting them uncritically.

Second, our findings illuminate the tension between *technological sophistication and human trust* in information systems. Studies of technology-mediated customer interactions reveal that while advanced digital interfaces can enhance service delivery, they may simultaneously introduce cognitive overload, distraction, and diminished trust if users cannot comprehend system logic (Pantano et al., 2024). This paradox extends to internal organizational systems: sophisticated AI classifiers may deliver superior technical performance yet fail to gain user acceptance if stakeholders perceive them as inscrutable black boxes. By generating interpretable recommendations, our system addresses this trust deficit, positioning explainability not as a luxury but as a prerequisite for effective information technology adoption.

Third, the agent agreement mechanism provides a practical solution to the *confidence calibration problem* ubiquitous in AI-driven information management. Organizations deploying automated systems must determine which decisions warrant full automation versus human review—a challenge particularly acute in domains where errors carry severe consequences. Our discovery that unanimous agent agreement predicts 92.28% accuracy while split decisions drop to 52.63% provides an actionable heuristic: agreement level serves as a reliable confidence signal for tiered automation architectures. This principle generalizes beyond document classification to any multi-model information system where diverse perspectives can be systematically aggregated and evaluated.

Extending these insights to emerging technological frontiers reveals both opportunities and challenges. The proliferation of immersive digital environments, including metaverse platforms and extended reality systems, will generate novel information management requirements as organizations navigate virtual workspaces, digital assets, and avatar-mediated communications (Dwivedi et al., 2022). Access control frameworks designed for traditional document repositories may require fundamental reconceptualization to address questions such as: How should sensitivity classifications apply to real-time virtual meetings where participants’ digital twins interact across jurisdictional boundaries? What explainability standards should govern AI agents that autonomously moderate access to virtual spaces or digital twins representing organizational assets? Can multi-agent architectures scale to handle the exponentially larger information flows characteristic of persistent, cross-platform virtual environments? While speculative, these questions underscore that the information management challenges we address—transparency, trust, and confidence calibration—will intensify rather than diminish as organizational information ecosystems become increasingly complex and technologically mediated. Our multi-agent framework provides a foundational architecture adaptable to these future contexts, demonstrating that explainable, collaborative AI can address information governance challenges across technological paradigms.

5.3 Ethical Considerations

5.3.1 Bias and Fairness

Our synthetic labeling methodology introduces potential biases through keyword selection and sentiment-based adjustments. If certain types of content systematically trigger higher sensitivity classifications due to keyword presence rather than actual sensitivity, the system could perpetuate these biases.

Furthermore, the models may have learned biases from their pre-training data. For example, if certain topics, companies, or geographies are over-represented in training corpora, classification patterns might reflect these imbalances. Careful validation and bias testing are essential before deployment.

5.3.2 Security and Adversarial Robustness

Access control systems are potential targets for adversarial attacks. Sophisticated actors might craft documents designed to evade high-sensitivity classification, enabling unauthorized access to restricted information. Our keyword-based approach is particularly vulnerable to keyword avoidance strategies. While we have not evaluated adversarial robustness, this represents a critical consideration for production deployment. Defense strategies might include adversarial training with deliberately obfuscated texts, semantic analysis beyond keyword matching, human review for unusual classification patterns, and regular model updating to address emerging evasion techniques.

5.3.3 Accountability and Human Oversight

While our system generates explanations, critical questions remain about accountability: Who is responsible when an incorrect classification leads to unauthorized disclosure or inappropriate access denial? How should human oversight be integrated? What training do human reviewers need to effectively validate system decisions?

Our work provides technical capability but does not address governance frameworks, liability allocation, or human-AI interaction protocols necessary for responsible deployment. Organizations implementing such systems must develop comprehensive governance structures alongside technical implementation. Studies of organizational responses to cybersecurity incidents underscore that technical controls alone are insufficient; effective information security requires integrated frameworks that address both technological safeguards and human factors, including organizational culture, management support, and employee training (Stacey et al., 2021). Similarly, regulatory frameworks for supply chain cybersecurity emphasize that compliance requires holistic approaches combining policy development, risk assessment, and stakeholder coordination (Gokkaya et al., 2025).

5.3.4 Organizational Implementation and Change Management

Deploying multi-agent AI systems for information access control introduces organizational change management

challenges that extend beyond technical integration. Financial institutions considering adoption must navigate resistance from information professionals whose expertise and judgment the system partially automates, legal and compliance teams requiring assurance that automated classifications meet regulatory standards, IT departments managing infrastructure for hosting multiple AI models and orchestrating their interactions, and executives evaluating return on investment against implementation costs and organizational disruption. Drawing on research in digital transformation and governance, risk, and compliance (GRC) system implementations (Spanaki & Papazafeiropoulou, 2016), we identify three critical success factors for organizational adoption.

First, *stakeholder alignment* requires early engagement with all affected parties to identify concerns, incorporate domain expertise into system design, and establish clear roles and responsibilities. Implementations that treat AI deployment as purely technical projects risk alienating stakeholders whose buy-in is essential for long-term success. Second, *phased rollout* strategies that begin with low-risk document categories and gradually expand to more sensitive classifications enable organizations to build confidence, refine orchestrator prompts based on real-world feedback, and identify edge cases requiring additional training data. Full-scale deployment without pilot testing increases the risk of high-profile failures that undermine organizational trust. Third, *continuous evaluation and adaptation* mechanisms that monitor classification accuracy, track override patterns and rationales, solicit user feedback on explanation quality, and update models periodically ensure that the system evolves alongside changing organizational needs and regulatory requirements. Research on digital transformation emphasizes that successful AI adoption requires viewing technology as enabler of organizational capabilities rather than standalone solution (Spanaki et al., 2024), a principle directly applicable to information access control automation.

5.4 Future Work

5.4.1 Real-World Dataset Validation

The highest priority for future work is validation on real financial institution access control datasets with ground-truth labels from domain experts. While confidentiality concerns make such datasets difficult to obtain, collaborations with financial institutions, anonymization techniques, or synthetic data generation based on real policy documents could enable more realistic evaluation. Successful FinTech implementations demonstrate that partnerships between academic researchers and industry practitioners

can yield valuable insights while respecting confidentiality constraints, particularly when projects are structured with clear success metrics and stakeholder engagement protocols (Jinasena et al., 2023).

5.4.2 Enhanced Orchestrator Training

Rather than relying on zero-shot reasoning, future work should explore fine-tuning the orchestrator on agent prediction patterns. A meta-learning approach could train the orchestrator specifically to synthesize agent outputs effectively, potentially improving override accuracy while maintaining explainability benefits. Advanced frameworks for orchestrator-agent trust dynamics, which model how reasoning systems should weight specialist opinions under uncertainty, could provide theoretical foundations for such meta-learning strategies (Roumeliotis et al., 2026).

5.4.3 Dynamic Agent Weighting

Our current system treats all agents equally. Future systems could implement dynamic weighting based on:

- Per-class agent expertise (e.g., FinBERT weighted higher for Confidential documents)
- Prediction confidence levels
- Historical accuracy on similar documents
- Document characteristics (length, vocabulary, structure)

Dynamic agent weighting, such as that proposed by Roumeliotis et al. (2026), might improve performance while retaining the multi-agent architecture's explainability and robustness benefits.

5.4.4 Expanded Agent Ensemble

Additional specialist agents could enhance system performance and coverage, such as legal language models for regulatory compliance aspects, named entity recognition models to identify sensitive entities (executives, unreleased products, acquisition targets), temporal models to account for time-sensitive information aging, and multi-modal models for documents containing tables, charts, or images.

5.4.5 Active Learning Integration

The agent agreement metric provides a natural active learning signal: split decisions and low-agreement predictions identify valuable samples for expert labeling. A production system could automatically flag these cases for human review, continuously improving model performance through targeted annotation.

5.4.6 Regulatory Compliance and Policy Relevance

The explainability capabilities demonstrated by our multi-agent system directly address emerging regulatory requirements for transparent AI decision-making, most notably the European Union's AI Act which entered into force in August 2024 (European Commission, 2026). Specifically, our system's natural language recommendations align with Article 13 (Transparency and provision of information to deployers) and Article 86 (Classification rules for general-purpose AI models), which mandate that high-risk AI systems provide clear information about their functionality, capabilities, and limitations to enable deployers to interpret outputs and use the system appropriately. Our orchestrator-generated explanations satisfy these requirements by articulating the reasoning process, referencing specialist agent opinions, identifying confidence levels through agreement patterns, and providing actionable recommendations for human decision-makers. Furthermore, the system's agent agreement tracking creates an auditable decision trail meeting Article 12's requirements for record-keeping and logging capabilities. Beyond the EU AI Act, our approach supports compliance with financial sector-specific regulations including the U.S. Securities and Exchange Commission's guidance on AI governance, the Basel Committee on Banking Supervision's principles for operational resilience requiring explainability in automated systems, and the UK Financial Conduct Authority's requirements for algorithmic accountability. Future work should systematically evaluate the system's compliance with these regulatory frameworks through expert legal review, implement standardized explainability reporting formats aligned with regulatory expectations, and develop automated compliance verification mechanisms that map system outputs to specific regulatory requirements. Such integration would position our multi-agent architecture as not merely a technical solution but a governance-aligned framework for responsible AI deployment in heavily regulated financial domains.

6 Conclusion

In an era where AI systems increasingly govern access to sensitive financial information, the imperative for transparency and accountability has never been greater. This work bridges the critical gap between high-performance AI and explainable decision-making by introducing a novel multi-agent agentic recommender system for financial document sensitivity classification.

Our system integrates three fine-tuned specialist models—FinBERT, BERT-base-uncased, and GPT-4.1-mini—with a GPT-5.1 reasoning orchestrator operating in zero-shot

mode. This architecture achieves 83.71% overall accuracy while generating natural language explanations for every classification, surpassing traditional black-box approaches that prioritize performance at the expense of interpretability.

The cornerstone of our contribution lies in the discovery that agent agreement serves as a powerful confidence calibration mechanism. When all three agents unanimously agree—representing 78.8% of cases—the system attains 92.28% accuracy, significantly outperforming any individual model. This insight enables tiered automation where high-confidence predictions can be fully automated, while split decisions trigger human oversight, creating a robust framework for responsible AI deployment in regulated financial environments.

Beyond technical innovation, our work establishes a blueprint for trustworthy AI in high-stakes domains. By providing interpretable recommendations that justify access-control decisions, the system addresses regulatory mandates for transparency, enables audit trails for compliance, and builds stakeholder trust essential for widespread AI adoption.

As artificial intelligence continues to reshape critical decision-making processes, our multi-agent approach demonstrates that explainability and performance are not mutually exclusive. This work paves the way for accountable AI systems that augment human judgment rather than replace it, fostering a future where advanced technology serves human values, institutional integrity, and societal well-being.

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Data Availability The Financial PhraseBank dataset is publicly available from Hugging Face (Malo et al., 2014). Our processed datasets, including the synthetically generated sensitivity labels, are available on GitHub (Applied-AI-Research-Lab, 2025).

Materials availability Not applicable.

Code availability The complete source code for this work, including data preparation scripts, model fine-tuning code, agentic system implementation, and evaluation tools, is publicly available as open-source on GitHub to support reproducibility and enable community extensions (Applied-AI-Research-Lab, 2025).

Declarations

Conflict of interest/Competing interests: The authors declare that they have no competing interests or conflicts of interest related to this work.

Ethics approval and consent to participate Not applicable. This research uses publicly available datasets and does not involve human subjects.

Consent for publication Not applicable.

Statement on AI Writing Assistance ChatGPT and Grammarly were utilized to enhance grammatical accuracy and refine sentence structure; all AI-generated revisions were thoroughly reviewed and edited for relevance.

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